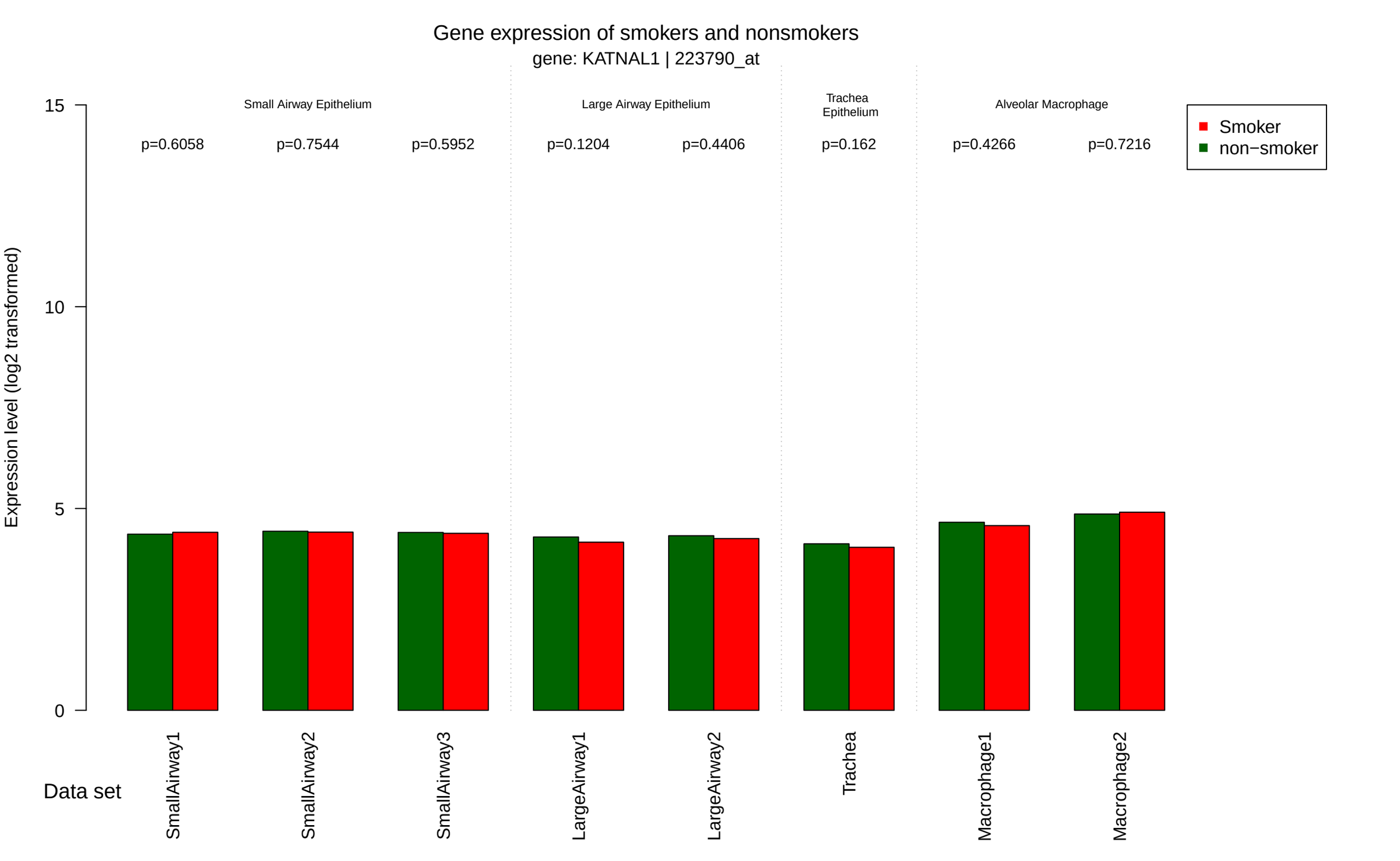
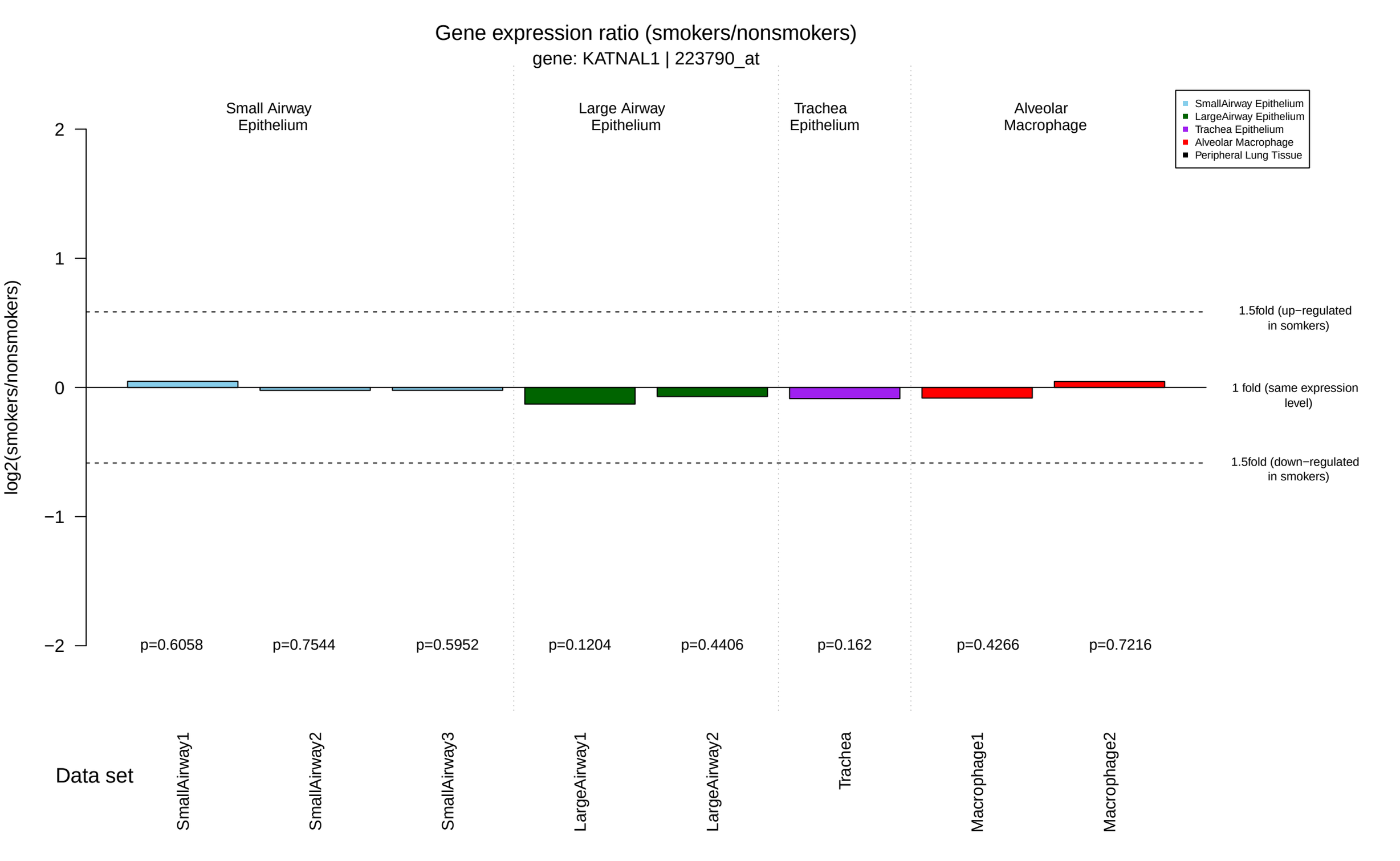
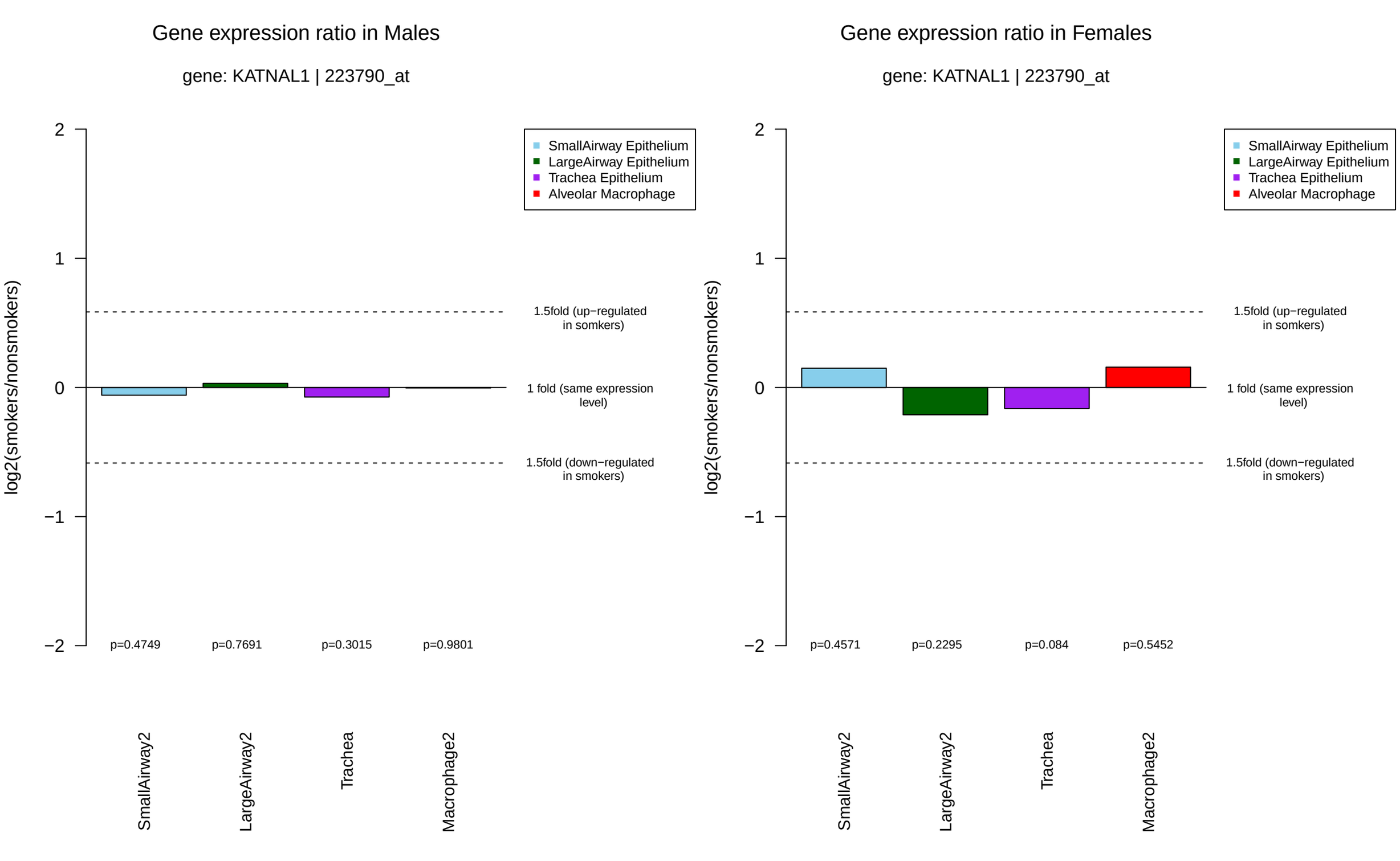
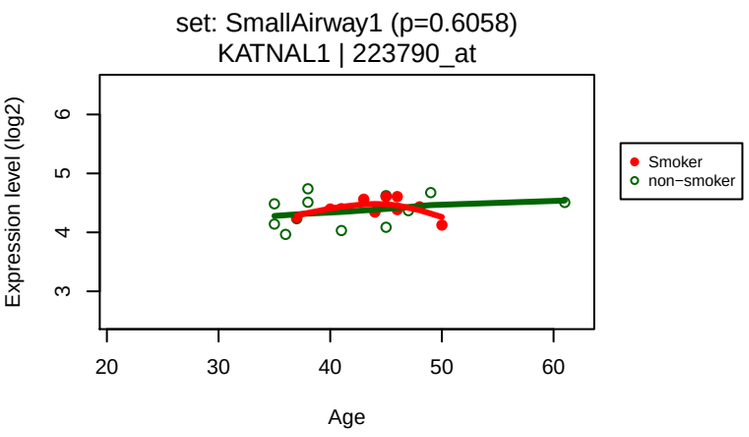
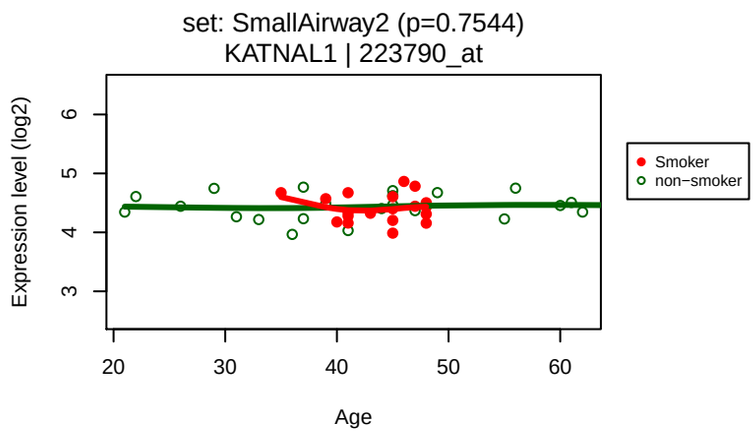
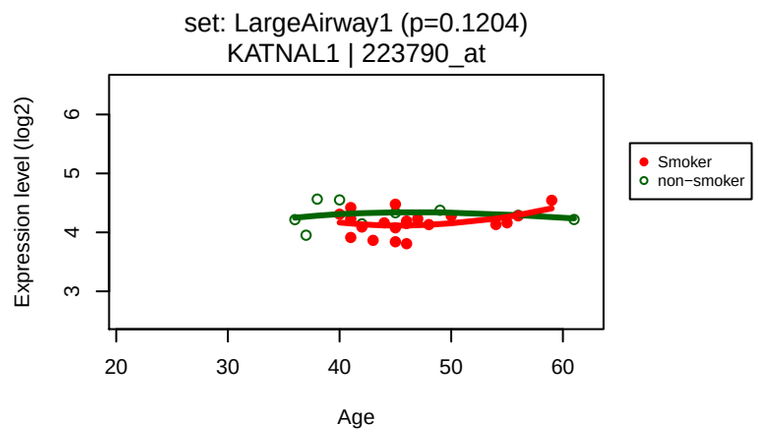

Gene expression of smokers and nonsmokers
gene: KATNAL1 | 223790_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.6058
p=0.7544
p=0.5952
p=0.1204
p=0.4406
p=0.162
p=0.4266
p=0.7216
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

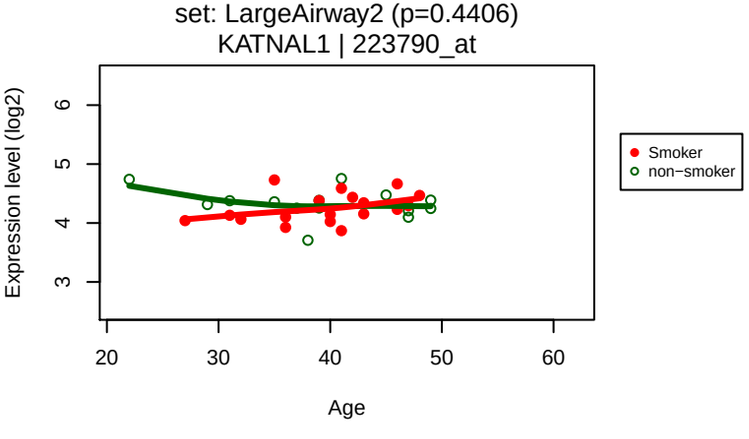

Gene expression ratio (smokers/nonsmokers)
gene: KATNAL1 | 223790_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6058
p=0.7544
p=0.5952
p=0.1204
p=0.4406
p=0.162
p=0.4266
p=0.7216
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

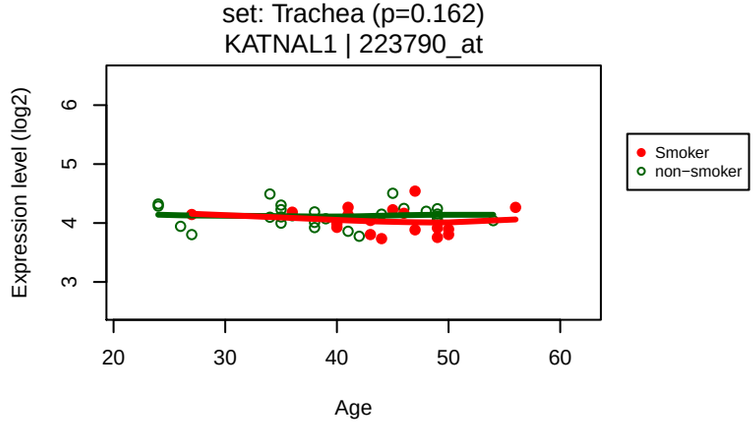

Gene expression ratio in Males
gene: KATNAL1 | 223790_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4749
p=0.7691
p=0.3015
p=0.9801
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: KATNAL1 | 223790_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4571
p=0.2295
p=0.084
p=0.5452
SmallAirway2
LargeAirway2
Trachea
Macrophage2

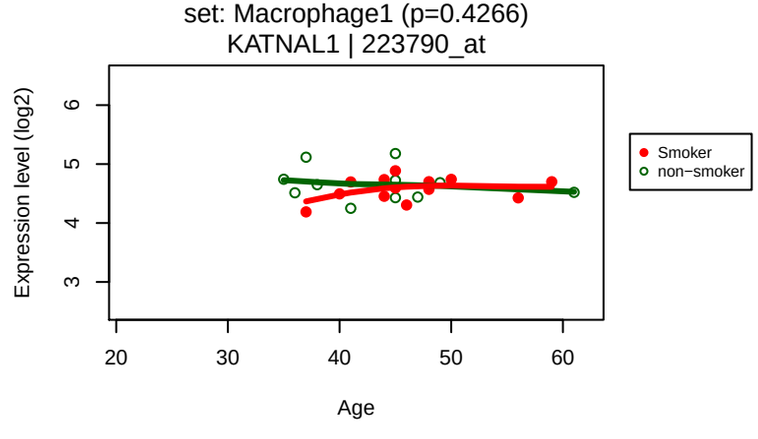

set: SmallAirway1 (p=0.6058)
KATNAL1 | 223790_at
Expression level (log2)
Age
Smoker
non−smoker

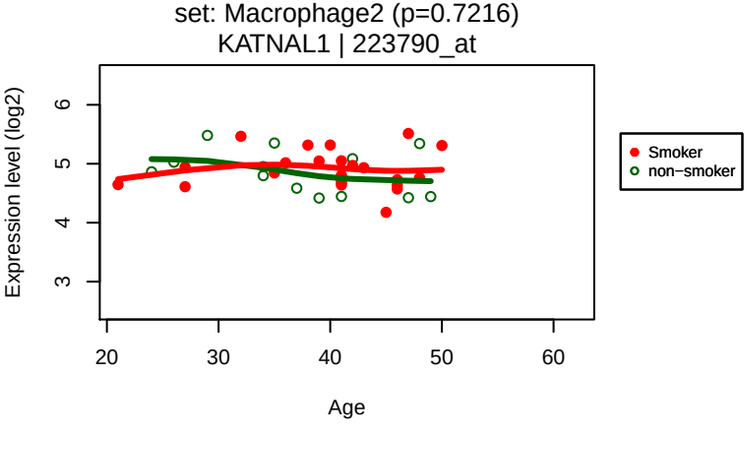

set: SmallAirway2 (p=0.7544)
KATNAL1 | 223790_at
Expression level (log2)
Age
Smoker
non−smoker

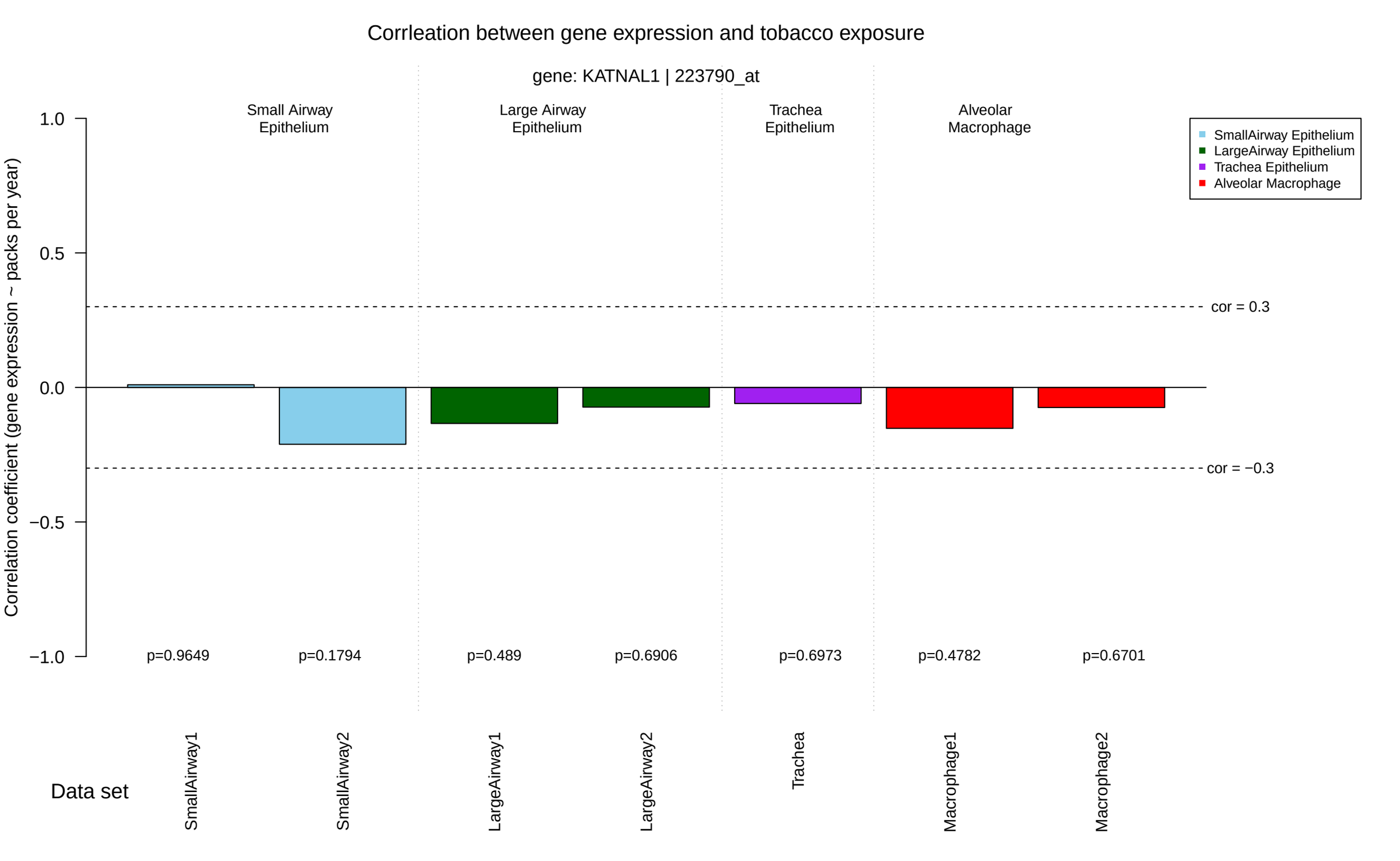

set: LargeAirway1 (p=0.1204)
KATNAL1 | 223790_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.4406)
KATNAL1 | 223790_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.162)
KATNAL1 | 223790_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4266)
KATNAL1 | 223790_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7216)
KATNAL1 | 223790_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: KATNAL1 | 223790_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.9649
p=0.1794
p=0.489
p=0.6906
p=0.6973
p=0.4782
p=0.6701
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2